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Housing Price Analysis Using Multiple Linear Regression

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1 Project Overview

This project investigates the relationship between housing characteristics and residential sale prices using the Ames Housing dataset. The objective is to understand how structural and neighborhood characteristics are associated with housing prices by applying multiple linear regression. The analysis includes data cleaning, exploratory data analysis, model building, model diagnostics, response transformation, model selection, and interpretation of the final model.

2 Data Source

The dataset used in this project comes from the **AmesHousing** package in R. The package provides a cleaned version of the Ames Housing dataset originally described by De Cock (2011). The dataset contains information on **2,930 residential properties** located in Ames, Iowa, with **82 variables** describing housing characteristics and sale prices. In this project, the dataset is loaded using the `make_ames()` function and then further processed for regression analysis.

Reference:

De Cock, D. (2011). *Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project.* Journal of Statistics Education, 19(3).

3 1. Import Dataset

```
# Load the original Ames Housing dataset
housing_raw <- make_ames()

# Select variables used in this analysis
housing_data <- housing_raw %>%
  transmute(
    sale_price       = Sale_Price,
    living_area      = Gr_Liv_Area,
    basement_area    = Total_Bsmt_SF,
    overall_quality = Overall_Qual,
    year_built       = Year_Built,
    neighborhood     = Neighborhood,
    garage_capacity = Garage_Cars
  )

# Display the first six observations
head(housing_data)
```

```
## # A tibble: 6 × 7
##   sale_price living_area basement_area overall_quality
year_built neighborhood
##        <int>       <int>         <dbl> <fct>
<int> <fct>
## 1     215000        1656          1080 Above_Average
1960 North_Ames
## 2     105000         896           882 Average
1961 North_Ames
## 3     172000        1329          1329 Above_Average
1958 North_Ames
## 4     244000        2110          2110 Good
1968 North_Ames
## 5     189900        1629           928 Average
1997 Gilbert
## 6     195500        1604           926 Above_Average
1998 Gilbert
## # ℹ 1 more variable: garage_capacity <dbl>
```

```
# Display the dimensions of the dataset
dim(housing_data)
```

```
## [1] 2930    7
```

```
# Check the structure of the dataset
str(housing_data)
```

```
## tibble [2,930 × 7] (S3: tbl_df/tbl/data.frame)
##  $ sale_price     : int [1:2930] 215000 105000 172000
244000 189900 195500 213500 191500 236500 189000 ...
##  $ living_area    : int [1:2930] 1656 896 1329 2110 16
29 1604 1338 1280 1616 1804 ...
##  $ basement_area  : num [1:2930] 1080 882 1329 2110 92
8 ...
##  $ overall_quality: Factor w/ 10 levels "Very_Poor","P
oor",..: 6 5 6 7 5 6 8 8 8 7 ...
##  $ year_built     : int [1:2930] 1960 1961 1958 1968 1
997 1998 2001 1992 1995 1999 ...
##  $ neighborhood   : Factor w/ 29 levels "North_Ames","
College_Creek",..: 1 1 1 1 7 7 17 17 17 7 ...
##  $ garage_capacity: num [1:2930] 2 1 1 2 2 2 2 2 2 2
...
##  - attr(*, "spec")=List of 2
##   ..$ cols   :List of 82
##   .. ..$ Order          : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ PID            : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ MS SubClass    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ MS Zoning      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Lot Frontage   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Lot Area       : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Street         : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Alley          : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Lot Shape      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Land Contour   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Utilities      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Lot Config     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Land Slope     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Neighborhood   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Condition 1    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Condition 2    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Bldg Type      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ House Style    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Overall Qual   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Overall Cond   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Year Built     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Year Remod/Add : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Roof Style     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Roof Matl      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Exterior 1st   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Exterior 2nd   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Mas Vnr Type   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Mas Vnr Area   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Exter Qual     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Exter Cond     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Foundation     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Bsmt Qual      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Bsmt Cond      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Bsmt Exposure  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ BsmtFin Type 1 : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ BsmtFin SF 1   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ BsmtFin Type 2 : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ BsmtFin SF 2   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Bsmt Unf SF    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Total Bsmt SF  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Heating        : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Heating QC     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Central Air    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Electrical     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ 1st Flr SF     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ 2nd Flr SF     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Low Qual Fin SF: list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Gr Liv Area    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Bsmt Full Bath : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Bsmt Half Bath : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Full Bath      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Half Bath      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Bedroom AbvGr  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Kitchen AbvGr  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Kitchen Qual   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ TotRms AbvGrd  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Functional     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Fireplaces     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Fireplace Qu   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Garage Type    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Garage Yr Blt  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Garage Finish  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Garage Cars    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Garage Area    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Garage Qual    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Garage Cond    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Paved Drive    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Wood Deck SF   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Open Porch SF  : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Enclosed Porch : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ 3Ssn Porch     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Screen Porch   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Pool Area      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Pool QC        : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Fence          : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Misc Feature   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Misc Val       : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Mo Sold        : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Yr Sold        : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   .. ..$ Sale Type      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ Sale Condition : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_cha
racter" "collector"
##   .. ..$ SalePrice      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_int
eger" "collector"
##   ..$ default: list()
##   .. ..- attr(*, "class")= chr [1:2] "collector_guess"
"collector"
##   ..- attr(*, "class")= chr "col_spec"
```

4 2. Data Cleaning

```
# Check the dimensions of the dataset
dim(housing_data)
```

```
## [1] 2930    7
```

```
# Check missing values
missing_summary <- housing_data %>%
  summarise(across(everything(), ~ sum(is.na(.))))

kable(
  missing_summary,
  caption = "Number of Missing Values in Each Variable"
)
```

Number of Missing Values in Each Variable

sale_price	living_area	basement_area	overall_quality	year_built	neighborhood	garage_capacity
0	0	0	0	0	0	0

```
# Remove observations with missing values
housing_clean <- housing_data %>%
  drop_na()

# Compare sample size before and after cleaning
data.frame(
  Dataset = c("Original", "Cleaned"),
  Observations = c(nrow(housing_data), nrow(housing_clea
n))
) %>%
  kable(
    caption = "Sample Size Before and After Data Cleanin
g"
  )
```

Sample Size Before and After Data Cleaning

Dataset	Observations
Original	2930
Cleaned	2930

5 3. Exploratory Data Analysis

5.1 3.1 Response Variable

5.1.1 Summary Statistics

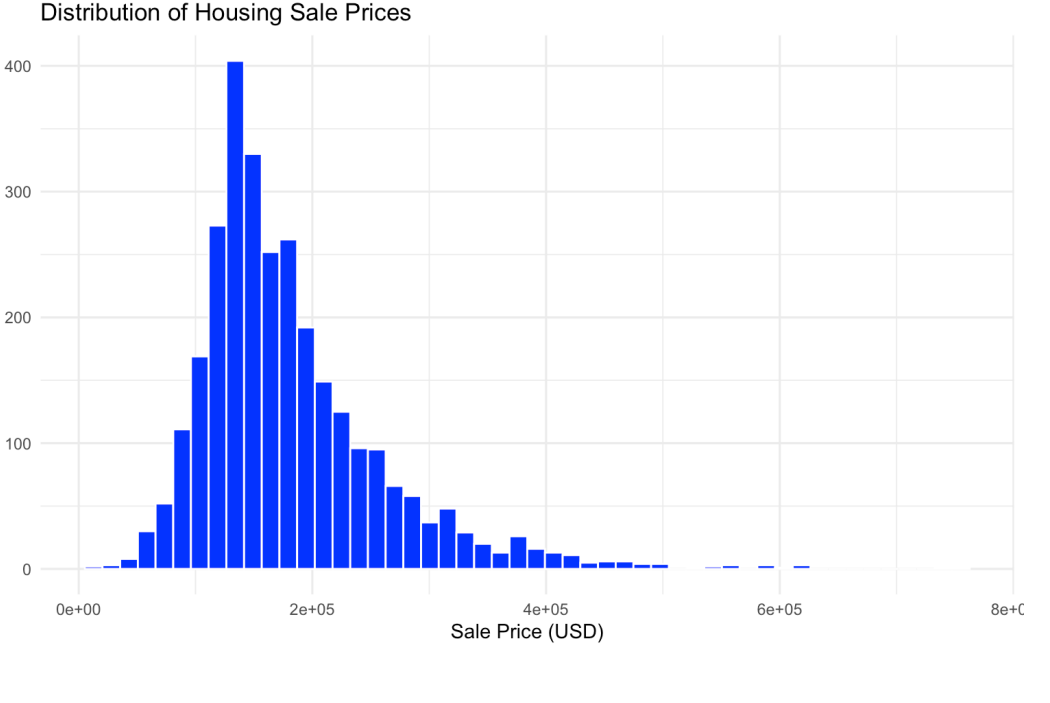
```
sum_price <- summary(housing_clean$sale_price)

tbl_price <- data.frame(
  Statistic = c(
    "Minimum",
    "First Quartile",
    "Median",
    "Mean",
    "Third Quartile",
    "Maximum",
    "Standard Deviation",
    "Interquartile Range"
  ),
  Value = c(
    sum_price["Min."],
    sum_price["1st Qu."],
    sum_price["Median"],
    sum_price["Mean"],
    sum_price["3rd Qu."],
    sum_price["Max."],
    sd(housing_clean$sale_price),
    IQR(housing_clean$sale_price)
  )
)

kable(
  tbl_price,
  digits = 2,
  caption = "Summary Statistics for Housing Sale Price"
)
```

Summary Statistics for Housing Sale Price

Statistic	Value
Minimum	12789.00
First Quartile	129500.00
Median	160000.00
Mean	180796.06
Third Quartile	213500.00
Maximum	755000.00
Standard Deviation	79886.69
Interquartile Range	84000.00

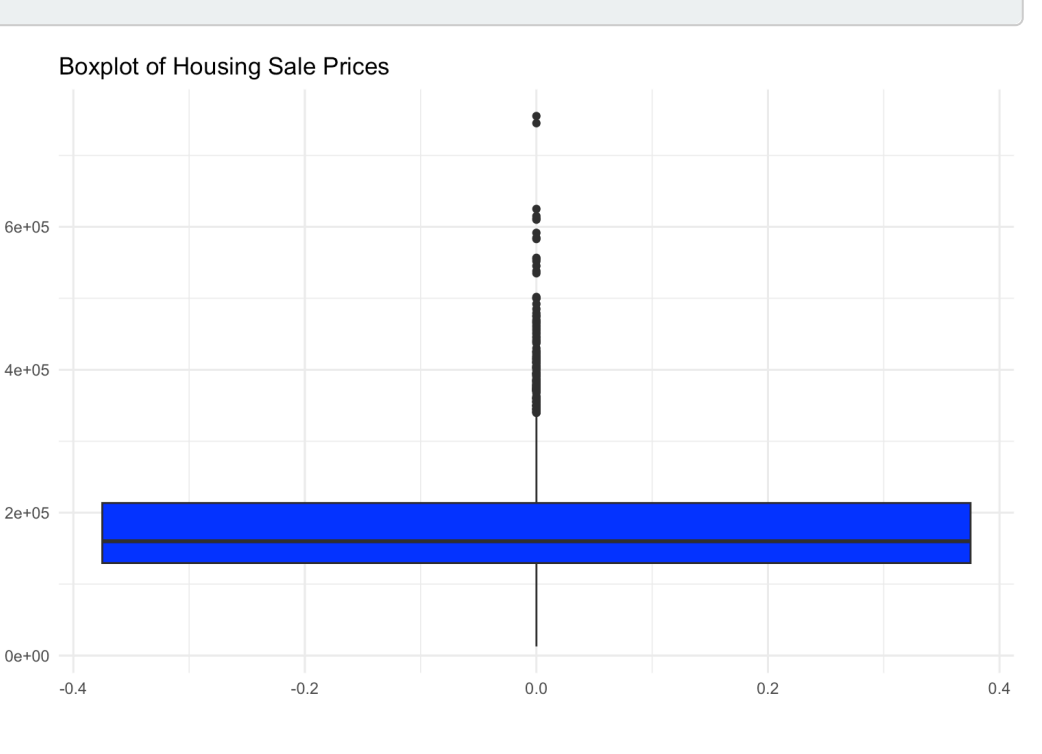
5.1.2 Distribution of Housing Sale Price

```
ggplot(housing_clean, aes(x = sale_price)) +
  geom_histogram(
    bins = 50,
    fill = "blue",
    color = "white"
  ) +
  labs(
    title = "Distribution of Housing Sale Prices",
    x = "Sale Price (USD)",
    y = "Frequency"
  ) +
  theme_minimal()
```



5.1.3 Boxplot of Housing Sale Price

```
ggplot(housing_clean, aes(y = sale_price)) +
  geom_boxplot(
    fill = "blue"
  ) +
  labs(
    title = "Boxplot of Housing Sale Prices",
    y = "Sale Price (USD)"
  ) +
  theme_minimal()
```



5.2 3.2 Numerical Predictors

5.2.1 Summary Statistics

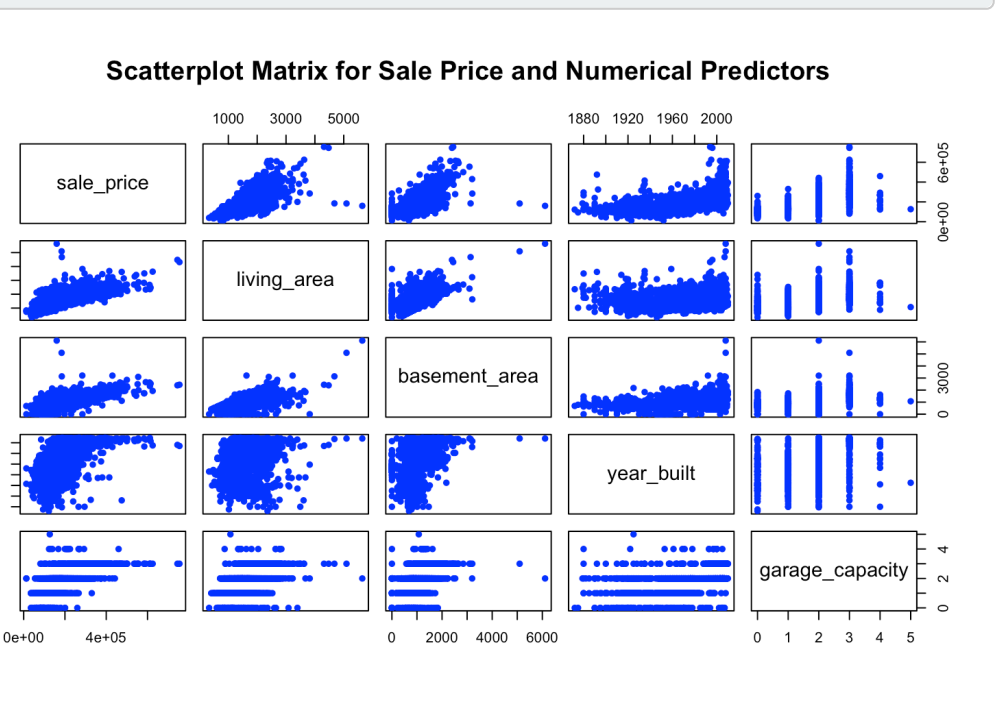
```
num_summary <- data.frame(
  Variable = c(
    "Living Area",
    "Basement Area",
    "Year Built",
    "Garage Capacity"
  ),
  Mean = c(
    mean(housing_clean$living_area),
    mean(housing_clean$basement_area),
    mean(housing_clean$year_built),
    mean(housing_clean$garage_capacity)
  ),
  SD = c(
    sd(housing_clean$living_area),
    sd(housing_clean$basement_area),
    sd(housing_clean$year_built),
    sd(housing_clean$garage_capacity)
  ),
  Min = c(
    min(housing_clean$living_area),
    min(housing_clean$basement_area),
    min(housing_clean$year_built),
    min(housing_clean$garage_capacity)
  ),
  Max = c(
    max(housing_clean$living_area),
    max(housing_clean$basement_area),
    max(housing_clean$year_built),
    max(housing_clean$garage_capacity)
  )
)

kable(
  num_summary,
  digits = 2,
  caption = "Summary Statistics for Numerical Predictors"
)
```

Summary Statistics for Numerical Predictors

Variable	Mean	SD	Min	Max
Living Area	1499.69	505.51	334	5642
Basement Area	1051.26	440.97	0	6110
Year Built	1971.36	30.25	1872	2010
Garage Capacity	1.77	0.76	0	5

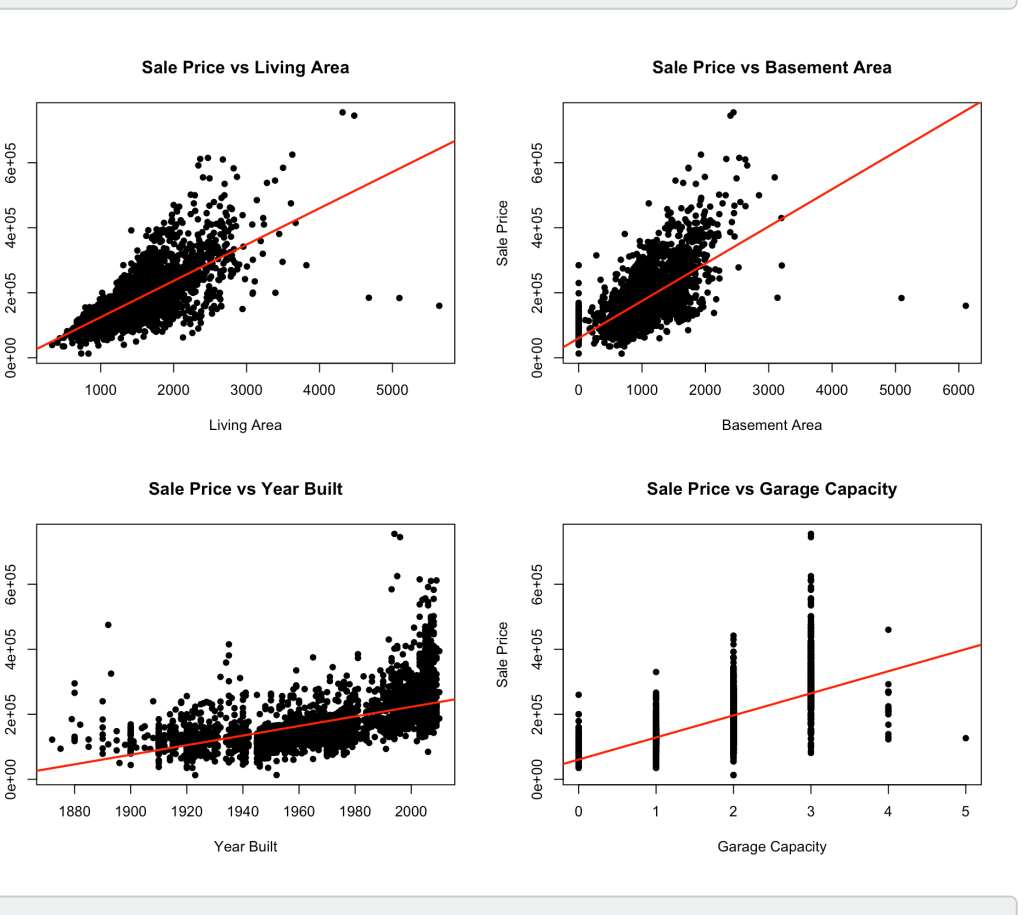
5.2.2 Scatterplot Matrix

```
pairs(
  housing_clean[, c(
    "sale_price",
    "living_area",
    "basement_area",
    "year_built",
    "garage_capacity"
  )],
  pch = 16,
  col = "blue",
  main = "Scatterplot Matrix for Sale Price and Numerical
Predictors"
)
```



5.2.3 Relationships Between Sale Price and Numerical Predictors

```
par(mfrow = c(2,2))

plot(
  housing_clean$living_area,
  housing_clean$sale_price,
  main = "Sale Price vs Living Area",
  xlab = "Living Area",
  ylab = "Sale Price",
  pch = 16,
  col = "black"
)
abline(
  lm(sale_price ~ living_area, data = housing_clean),
  col = "red",
  lwd = 2
)

plot(
  housing_clean$basement_area,
  housing_clean$sale_price,
  main = "Sale Price vs Basement Area",
  xlab = "Basement Area",
  ylab = "Sale Price",
  pch = 16,
  col = "black"
)
abline(
  lm(sale_price ~ basement_area, data = housing_clean),
  col = "red",
  lwd = 2
)

plot(
  housing_clean$year_built,
  housing_clean$sale_price,
  main = "Sale Price vs Year Built",
  xlab = "Year Built",
  ylab = "Sale Price",
  pch = 16,
  col = "black"
)
abline(
  lm(sale_price ~ year_built, data = housing_clean),
  col = "red",
  lwd = 2
)

plot(
  housing_clean$garage_capacity,
  housing_clean$sale_price,
  main = "Sale Price vs Garage Capacity",
  xlab = "Garage Capacity",
  ylab = "Sale Price",
  pch = 16,
  col = "black"
)
abline(
  lm(sale_price ~ garage_capacity, data = housing_clean),
  col = "red",
  lwd = 2
)
```

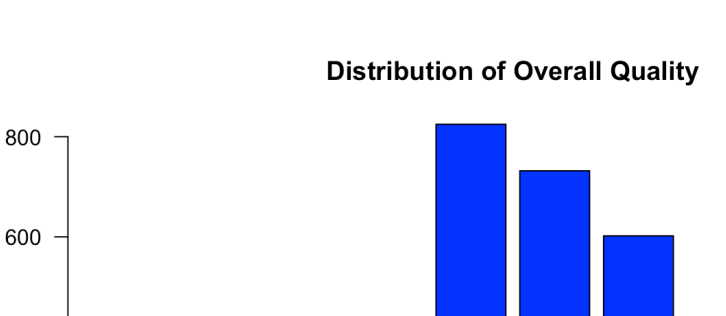


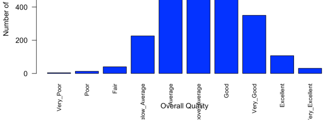
```
par(mfrow = c(1,1))
```

5.3 3.3 Categorical Predictors

5.3.1 Distribution of Overall Quality

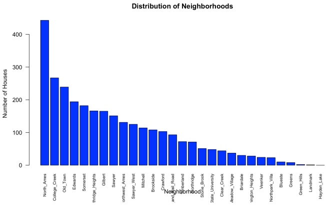
```
barplot(
  table(housing_clean$overall_quality),
  main = "Distribution of Overall Quality",
  xlab = "Overall Quality",
  ylab = "Number of Houses",
  col = "blue",
  las = 2,
  cex.names = 0.8
)
```





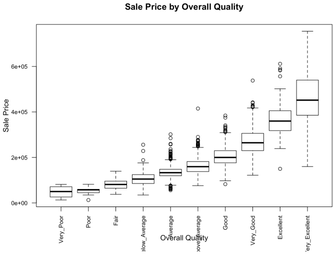
5.3.2 Distribution of Neighborhoods

```
barplot(
  table(housing_clean$neighborhood),
  main = "Distribution of Neighborhoods",
  xlab = "Neighborhood",
  ylab = "Number of Houses",
  col = "blue",
  las = 2,
  cex.names = 0.7
)
```



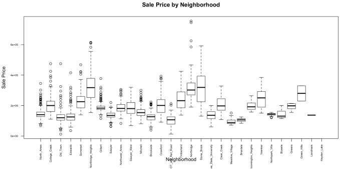
5.3.3 Sale Price by Overall Quality

```
boxplot(
  sale_price ~ overall_quality,
  data = housing_clean,
  main = "Sale Price by Overall Quality",
  xlab = "Overall Quality",
  ylab = "Sale Price",
  col = "white",
  las = 2,
  cex.axis = 0.8
)
```



5.3.4 Sale Price by Neighborhood

```
boxplot(
  sale_price ~ neighborhood,
  data = housing_clean,
  main = "Sale Price by Neighborhood",
  xlab = "Neighborhood",
  ylab = "Sale Price",
  col = "white",
  las = 2,
  cex.axis = 0.55
)
```



6 4. Initial Regression Model

6.1 4.1 Multiple Linear Regression Model

```
initial_model <- lm(
  sale_price ~
    living_area +
    basement_area +
    overall_quality +
    year_built +
    neighborhood +
    garage_capacity,
  data = housing_clean
)

summary(initial_model)
```

```
##
## Call:
## lm(formula = sale_price ~ living_area + basement_area
+ overall_quality +
##     year_built + neighborhood + garage_capacity, data
= housing_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -453609  -13100     519   12449  213165
##
## Coefficients:
##
Estimate Std. Error
## (Intercept)                                          -
6.180e+05  8.658e+04
## living_area
4.613e+01  1.613e+00
## basement_area
1.985e+01  1.777e+00
## overall_qualityPoor
1.640e+04  1.797e+04
## overall_qualityFair
2.295e+04  1.647e+04
## overall_qualityBelow_Average
3.118e+04  1.588e+04
## overall_qualityAverage
4.030e+04  1.580e+04
## overall_qualityAbove_Average
4.847e+04  1.584e+04
## overall_qualityGood
6.156e+04  1.593e+04
## overall_qualityVery_Good
9.057e+04  1.610e+04
## overall_qualityExcellent
1.578e+05  1.646e+04
## overall_qualityVery_Excellent
1.986e+05  1.733e+04
## year_built
3.182e+02  4.359e+01
## neighborhoodCollege_Creek
7.063e+03  2.978e+03
## neighborhoodOld_Town                                 -
9.918e+03  3.015e+03
## neighborhoodEdwards                                  -
8.730e+03  2.772e+03
## neighborhoodSomerset
9.926e+03  3.520e+03
## neighborhoodNorthridge_Heights
3.491e+04  3.909e+03
## neighborhoodGilbert
1.843e+03  3.424e+03
```

```
## neighborhoodSawyer
5.757e+01  2.979e+03
## neighborhoodNorthwest_Ames
1.237e+03  3.294e+03
## neighborhoodSawyer_West
7.773e+01  3.497e+03
## neighborhoodMitchell
1.020e+03  3.423e+03
## neighborhoodBrookside
1.765e+03  3.580e+03
## neighborhoodCrawford
3.283e+04  3.591e+03
## neighborhoodIowa_DOT_and_Rail_Road                   -
1.003e+04  3.899e+03
## neighborhoodTimberland
2.020e+04  4.371e+03
## neighborhoodNorthridge
4.733e+04  4.728e+03
## neighborhoodStone_Brook
4.378e+04  5.317e+03
## neighborhoodSouth_and_West_of_Iowa_State_University -
8.137e+03  4.986e+03
## neighborhoodClear_Creek
2.748e+04  5.015e+03
## neighborhoodMeadow_Village                           -
1.805e+04  5.661e+03
## neighborhoodBriardale                                -
2.477e+04  6.048e+03
## neighborhoodBloomington_Heights                      -
4.396e+03  6.513e+03
## neighborhoodVeenker
2.368e+04  6.744e+03
## neighborhoodNorthpark_Villa                          -
1.712e+04  6.802e+03
## neighborhoodBlueste                                  -
1.920e+04  1.015e+04
## neighborhoodGreens                                   -
5.682e+03  1.149e+04
## neighborhoodGreen_Hills
1.126e+05  2.232e+04
## neighborhoodLandmark                                 -
2.223e+04  3.141e+04
## garage_capacity
1.058e+04  1.078e+03
##                                                      t
value Pr(>|t|)
## (Intercept)                                           -
7.137 1.20e-12 ***
## living_area                                           2
8.594  < 2e-16 ***
## basement_area                                         1
1.176  < 2e-16 ***
## overall_qualityPoor
0.913 0.361574
## overall_qualityFair
1.393 0.163693
## overall_qualityBelow_Average
1.963 0.049694 *
## overall_qualityAverage
2.550 0.010828 *
## overall_qualityAbove_Average
3.060 0.002233 **
## overall_qualityGood
3.866 0.000113 ***
## overall_qualityVery_Good
5.627 2.02e-08 ***
## overall_qualityExcellent
9.583  < 2e-16 ***
## overall_qualityVery_Excellent                         1
1.460  < 2e-16 ***
## year_built
7.300 3.69e-13 ***
## neighborhoodCollege_Creek
2.372 0.017750 *
## neighborhoodOld_Town                                  -
3.290 0.001014 **
## neighborhoodEdwards                                   -
3.150 0.001650 **
## neighborhoodSomerset
2.820 0.004839 **
## neighborhoodNorthridge_Heights
8.930  < 2e-16 ***
## neighborhoodGilbert
0.538 0.590489
## neighborhoodSawyer
0.019 0.984582
```

```
## neighborhoodNorthwest_Ames
0.375 0.707319
## neighborhoodSawyer_West
0.022 0.982271
## neighborhoodMitchell
0.298 0.765821
## neighborhoodBrookside
0.493 0.622029
## neighborhoodCrawford
9.142  < 2e-16 ***
## neighborhoodIowa_DOT_and_Rail_Road                   -
2.573 0.010145 *
## neighborhoodTimberland
4.621 3.99e-06 ***
## neighborhoodNorthridge                               1
0.011  < 2e-16 ***
## neighborhoodStone_Brook
8.234 2.72e-16 ***
## neighborhoodSouth_and_West_of_Iowa_State_University  -
1.632 0.102811
## neighborhoodClear_Creek
5.480 4.62e-08 ***
## neighborhoodMeadow_Village                           -
3.188 0.001447 **
## neighborhoodBriardale                                -
4.096 4.31e-05 ***
## neighborhoodBloomington_Heights                      -
0.675 0.499823
## neighborhoodVeenker
3.511 0.000453 ***
## neighborhoodNorthpark_Villa                          -
2.517 0.011903 *
## neighborhoodBlueste                                  -
1.891 0.058709 .
## neighborhoodGreens                                   -
0.494 0.621036
## neighborhoodGreen_Hills
5.043 4.85e-07 ***
## neighborhoodLandmark                                 -
0.708 0.479136
## garage_capacity
9.812  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1
##
## Residual standard error: 31320 on 2889 degrees of free
dom
## Multiple R-squared:  0.8484, Adjusted R-squared:  0.84
63
## F-statistic: 404.3 on 40 and 2889 DF,  p-value: < 2.2e
-16
```

6.2 4.2 Initial Model Coefficients

```
initial_coefficient_table <- broom::tidy(
  initial_model,
  conf.int = TRUE,
  conf.level = 0.95
)

initial_coefficient_table <- initial_coefficient_table[,
c(
  "term",
  "estimate",
  "std.error",
  "statistic",
  "p.value",
  "conf.low",
  "conf.high"
)]

colnames(initial_coefficient_table) <- c(
  "Variable",
  "Estimate",
  "Standard Error",
  "t Statistic",
  "p-value",
  "95% CI Lower",
  "95% CI Upper"
)

kable(
  initial_coefficient_table,
```

```
  digits = 4,
  caption = "Coefficient Estimates for the Initial Regression Model"
)
```

Coefficient Estimates for the Initial Regression Model

Variable	Estimate	Standard Error	t Statistic	p-value	95% CI Lower	95% CI Upper
(Intercept)	-617990.5316	86583.9375	-7.1375	0.0000	-787763.0574	-448218.0057
living_area	46.1330	1.6134	28.5942	0.0000	42.9696	49.2965
basement_area	19.8542	1.7766	11.1757	0.0000	16.3708	23.3377
overall_qualityPoor	16400.6276	17973.0051	0.9125	0.3616	-18840.5795	51641.8348
overall_qualityFair	22945.6364	16470.7621	1.3931	0.1637	-9349.9945	55241.2674
overall_qualityBelow_Average	31178.9733	15879.9720	1.9634	0.0497	41.7552	62316.1915
overall_qualityAverage	40296.2928	15803.4011	2.5498	0.0108	9309.2137	71283.3719
overall_qualityAbove_Average	48473.8315	15840.9705	3.0600	0.0022	17413.0869	79534.5761
overall_qualityGood	61561.3754	15925.5783	3.8656	0.0001	30334.7331	92788.0177
overall_qualityVery_Good	90567.2102	16096.5012	5.6265	0.0000	59005.4246	122128.9958
overall_qualityExcellent	157750.1685	16461.5892	9.5829	0.0000	125472.5237	190027.8132
overall_qualityVery_Excellent	198616.5457	17331.0032	11.4602	0.0000	164634.1666	232598.9248
year_built	318.2088	43.5881	7.3004	0.0000	232.7419	403.6757
neighborhoodCollege_Creek	7063.1246	2977.5143	2.3722	0.0177	1224.8578	12901.3913
neighborhoodOld_Town	-9917.9897	3014.5461	-3.2900	0.0010	-15828.8678	-4007.1116
neighborhoodEdwards	-8730.0150	2771.5714	-3.1498	0.0017	-14164.4718	-3295.5581
neighborhoodSomerset	9926.3921	3520.2998	2.8198	0.0048	3023.8394	16828.9448
neighborhoodNorthridge_Heights	34908.3923	3908.9283	8.9304	0.0000	27243.8226	42572.9621
neighborhoodGilbert	1842.5581	3423.6368	0.5382	0.5905	-4870.4591	8555.5753
neighborhoodSawyer	57.5733	2978.8977	0.0193	0.9846	-5783.4061	5898.5526
neighborhoodNorthwest_Ames	1236.7492	3293.6375	0.3755	0.7073	-5221.3673	7694.8657
neighborhoodSawyer_West	77.7257	3497.3991	0.0222	0.9823	-6779.9236	6935.3750
neighborhoodMitchell	1019.6330	3423.0377	0.2979	0.7658	-5692.2096	7731.4755
neighborhoodBrookside	1765.2572	3580.4236	0.4930	0.6220	-5255.1854	8785.6998
neighborhoodCrawford	32831.5386	3591.1986	9.1422	0.0000	25789.9687	39873.1085
neighborhoodIowa_DOT_and_Rail_Road	-10030.9081	3899.2152	-2.5725	0.0101	-17676.4326	-2385.3836
neighborhoodTimberland	20195.3883	4370.6297	4.6207	0.0000	11625.5212	28765.2555
neighborhoodNorthridge	47330.7180	4728.0016	10.0107	0.0000	38060.1211	56601.3149
neighborhoodStone_Brook	43776.9031	5316.9042	8.2335	0.0000	33351.5946	54202.2115
neighborhoodSouth_and_West_of_Iowa_State_University	-8137.2631	4986.4015	-1.6319	0.1028	-17914.5266	1640.0004
neighborhoodClear_Creek	27484.1440	5015.2817	5.4801	0.0000	17650.2525	37318.0354
neighborhoodMeadow_Village	-18048.1390	5660.7771	-3.1883	0.0014	-29147.7085	-6948.5694
neighborhoodBriardale	-24772.5264	6047.6098	-4.0963	0.0000	-36630.5917	-12914.4610
neighborhoodBloomington_Heights	-4395.6536	6513.4895	-0.6749	0.4998	-17167.2091	8375.9019
neighborhoodVeenker	23680.2391	6744.3427	3.5111	0.0005	10456.0300	36904.4481
neighborhoodNorthpark_Villa	-17118.7872	6802.3281	-2.5166	0.0119	-30456.6933	-3780.8811
neighborhoodBlueste	-19204.1970	10154.9528	-1.8911	0.0587	-39115.8807	707.4868
neighborhoodGreens	-5681.6403	11491.1531	-0.4944	0.6210	-28213.3263	16850.0456
neighborhoodGreen_Hills	112550.9511	22316.2048	5.0435	0.0000	68793.6612	156308.2410
neighborhoodLandmark	-22234.6923	31414.4616	-0.7078	0.4791	-83831.7118	39362.3272
garage_capacity	10578.7813	1078.1690	9.8118	0.0000	8464.7232	12692.8395

6.3 4.3 Initial Model Fit

```
initial_model_fit <- broom::glance(initial_model)

model_fit_table <- data.frame(
  Measure = c(
    "R-squared",
    "Adjusted R-squared",
    "Residual Standard Error",
    "F Statistic",
    "Model p-value",
    "AIC",
    "BIC"
  ),
  Value = c(
    initial_model_fit$r.squared,
    initial_model_fit$adj.r.squared,
    initial_model_fit$sigma,
    initial_model_fit$statistic,
    initial_model_fit$p.value,
    AIC(initial_model),
    BIC(initial_model)
  )
)

kable(
  model_fit_table,
  digits = 4,
  caption = "Model Fit Measures for the Initial Regressio
n Model"
)
```

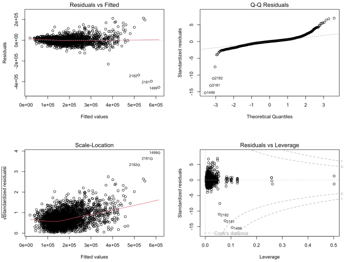
Model Fit Measures for the Initial Regression Model

Measure	Value
R-squared	0.8484
Adjusted R-squared	0.8463
Residual Standard Error	31317.2093
F Statistic	404.2522
Model p-value	0.0000
AIC	69019.9593
BIC	69271.2351

7 5. Initial Model Diagnostics

7.1 5.1 Standard Regression Diagnostic Plots

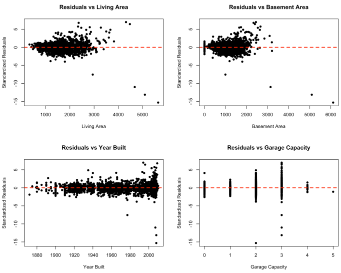
```
par(mfrow = c(2, 2))

plot(initial_model)
```



```
par(mfrow = c(1, 1))
```

7.2 5.2 Residual Plots for Numerical Predictors

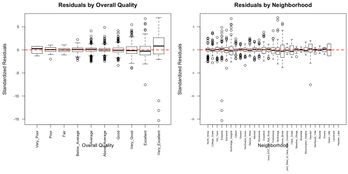
```r
initial_fitted_values <- fitted(initial_model)
initial_standardized_residuals <- rstandard(initial_model)

par(mfrow = c(2, 2))

plot(
  housing_clean$living_area,
  initial_standardized_residuals,
  main = "Residuals vs Living Area",
  xlab = "Living Area",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)

plot(
  housing_clean$basement_area,
  initial_standardized_residuals,
  main = "Residuals vs Basement Area",
  xlab = "Basement Area",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)

plot(
  housing_clean$year_built,
  initial_standardized_residuals,
  main = "Residuals vs Year Built",
  xlab = "Year Built",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)

plot(
  housing_clean$garage_capacity,
  initial_standardized_residuals,
  main = "Residuals vs Garage Capacity",
  xlab = "Garage Capacity",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)
```



```r
par(mfrow = c(1, 1))
```

7.3 5.3 Residual Plots for Categorical Predictors

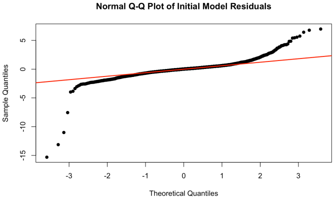
```
par(mfrow = c(1, 2), mar = c(8, 4, 3, 1))

boxplot(
  initial_standardized_residuals ~ housing_clean$overall_quality,
  main = "Residuals by Overall Quality",
  xlab = "Overall Quality",
  ylab = "Standardized Residuals",
  col = "white",
  las = 2,
  cex.axis = 0.75
)

abline(h = 0, col = "red", lty = 2, lwd = 2)

boxplot(
  initial_standardized_residuals ~ housing_clean$neighborhood,
  main = "Residuals by Neighborhood",
  xlab = "Neighborhood",
  ylab = "Standardized Residuals",
  col = "white",
  las = 2,
  cex.axis = 0.5
)

abline(h = 0, col = "red", lty = 2, lwd = 2)
```



```
par(mfrow = c(1, 1))
```

7.4 5.4 Normal Q-Q Plot

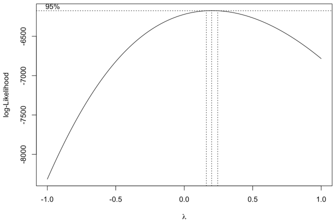
```
qqnorm(
  initial_standardized_residuals,
  main = "Normal Q-Q Plot of Initial Model Residuals",
  pch = 16,
  col = "black"
)

qqline(
  initial_standardized_residuals,
  col = "red",
  lwd = 2
)
```



8 6. Model Refinement

8.1 6.1 Box-Cox Transformation

```
boxcox_result <- MASS::boxcox(
  initial_model,
  lambda = seq(-1, 1, by = 0.01),
  plotit = TRUE
)
```



```
estimated_lambda <- boxcox_result$x[
  which.max(boxcox_result$y)
]

estimated_lambda
```

```
## [1] 0.2
```

8.2 6.2 Create the Transformed Response Variable

```
# Use a convenient value close to the estimated Box-Cox lambda
lambda_selected <- 0.25

housing_clean <- housing_clean %>%
  mutate(
    sale_price_bc =
      (sale_price^lambda_selected - 1) / lambda_selected
  )

head(
  housing_clean[, c(
    "sale_price",
    "sale_price_bc"
  )]
)
```

```
## # A tibble: 6 × 2
##   sale_price sale_price_bc
##        <int>         <dbl>
## 1     215000          82.1
## 2     105000          68.0
## 3     172000          77.5
## 4     244000          84.9
## 5     189900          79.5
## 6     195500          80.1
```

8.3 6.3 Transformed Multiple Linear Regression Model

```
transformed_model <- lm(
  sale_price_bc ~
    living_area +
    basement_area +
    overall_quality +
    year_built +
    neighborhood +
    garage_capacity,
  data = housing_clean
)

summary(transformed_model)
```

```
##
## Call:
## lm(formula = sale_price_bc ~ living_area + basement_ar
ea + overall_quality +
##     year_built + neighborhood + garage_capacity, data
= housing_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -40.334  -1.386   0.149   1.608  14.727
##
## Coefficients:
##
Estimate Std. Error
## (Intercept)                                         -
3.105e+01  8.550e+00
## living_area
4.911e-03  1.593e-04
## basement_area
2.082e-03  1.754e-04
## overall_qualityPoor
3.838e+00  1.775e+00
## overall_qualityFair
8.891e+00  1.626e+00
## overall_qualityBelow_Average
1.146e+01  1.568e+00
## overall_qualityAverage
1.360e+01  1.561e+00
## overall_qualityAbove_Average
1.496e+01  1.564e+00
## overall_qualityGood
1.644e+01  1.573e+00
## overall_qualityVery_Good
1.884e+01  1.590e+00
## overall_qualityExcellent
2.311e+01  1.626e+00
## overall_qualityVery_Excellent
2.331e+01  1.711e+00
## year_built
4.102e-02  4.304e-03
## neighborhoodCollege_Creek
6.965e-01  2.940e-01
## neighborhoodOld_Town                                -
1.525e+00  2.977e-01
## neighborhoodEdwards                                 -
1.280e+00  2.737e-01
## neighborhoodSomerset
1.056e+00  3.476e-01
## neighborhoodNorthridge_Heights
2.781e+00  3.860e-01
## neighborhoodGilbert
2.219e-01  3.381e-01
## neighborhoodSawyer                                  -
```

```
4.943e-02  2.942e-01
## neighborhoodNorthwest_Ames
2.183e-01  3.252e-01
## neighborhoodSawyer_West                             -
2.275e-01  3.454e-01
## neighborhoodMitchell
1.702e-01  3.380e-01
## neighborhoodBrookside                               -
2.512e-01  3.536e-01
## neighborhoodCrawford
3.594e+00  3.546e-01
## neighborhoodIowa_DOT_and_Rail_Road                  -
2.438e+00  3.850e-01
## neighborhoodTimberland
1.983e+00  4.316e-01
## neighborhoodNorthridge
3.043e+00  4.669e-01
## neighborhoodStone_Brook
3.161e+00  5.250e-01
## neighborhoodSouth_and_West_of_Iowa_State_University -
9.011e-01  4.924e-01
## neighborhoodClear_Creek
3.120e+00  4.953e-01
## neighborhoodMeadow_Village                          -
3.490e+00  5.590e-01
## neighborhoodBriardale                               -
4.514e+00  5.972e-01
## neighborhoodBloomington_Heights                     -
4.981e-01  6.432e-01
## neighborhoodVeenker
2.405e+00  6.660e-01
## neighborhoodNorthpark_Villa                         -
2.295e+00  6.717e-01
## neighborhoodBlueste                                 -
2.545e+00  1.003e+00
## neighborhoodGreens
2.457e-01  1.135e+00
## neighborhoodGreen_Hills
1.092e+01  2.204e+00
## neighborhoodLandmark                                -
3.062e+00  3.102e+00
## garage_capacity
1.278e+00  1.065e-01
##                                                     t
value Pr(>|t|)
## (Intercept)                                          -
3.632 0.000286 ***
## living_area                                          3
0.827  < 2e-16 ***
## basement_area                                        1
1.867  < 2e-16 ***
## overall_qualityPoor
2.162 0.030679 *
## overall_qualityFair
5.466 4.99e-08 ***
## overall_qualityBelow_Average
7.309 3.46e-13 ***
## overall_qualityAverage
8.717  < 2e-16 ***
## overall_qualityAbove_Average
9.566  < 2e-16 ***
## overall_qualityGood                                  1
0.455  < 2e-16 ***
## overall_qualityVery_Good                             1
1.855  < 2e-16 ***
## overall_qualityExcellent                             1
4.215  < 2e-16 ***
## overall_qualityVery_Excellent                        1
3.620  < 2e-16 ***
## year_built
9.530  < 2e-16 ***
## neighborhoodCollege_Creek
2.369 0.017907 *
## neighborhoodOld_Town                                 -
5.121 3.24e-07 ***
## neighborhoodEdwards                                  -
4.675 3.07e-06 ***
## neighborhoodSomerset
3.037 0.002407 **
## neighborhoodNorthridge_Heights
7.206 7.33e-13 ***
## neighborhoodGilbert
0.656 0.511607
## neighborhoodSawyer                                   -
0.168 0.866555
## neighborhoodNorthwest_Ames
```

```
0.671 0.502237
## neighborhoodSawyer_West                              -
0.659 0.510044
## neighborhoodMitchell
0.504 0.614650
## neighborhoodBrookside                                -
0.710 0.477459
## neighborhoodCrawford                                 1
0.135  < 2e-16 ***
## neighborhoodIowa_DOT_and_Rail_Road                   -
6.331 2.81e-10 ***
## neighborhoodTimberland
4.595 4.52e-06 ***
## neighborhoodNorthridge
6.517 8.45e-11 ***
## neighborhoodStone_Brook
6.021 1.95e-09 ***
## neighborhoodSouth_and_West_of_Iowa_State_University  -
1.830 0.067366 .
## neighborhoodClear_Creek
6.299 3.45e-10 ***
## neighborhoodMeadow_Village                           -
6.243 4.92e-10 ***
## neighborhoodBriardale                                -
7.559 5.41e-14 ***
## neighborhoodBloomington_Heights                      -
0.774 0.438782
## neighborhoodVeenker
3.611 0.000310 ***
## neighborhoodNorthpark_Villa                          -
3.416 0.000644 ***
## neighborhoodBlueste                                  -
2.538 0.011199 *
## neighborhoodGreens
0.217 0.828602
## neighborhoodGreen_Hills
4.956 7.62e-07 ***
## neighborhoodLandmark                                 -
0.987 0.323765
## garage_capacity                                      1
2.001  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1
##
## Residual standard error: 3.093 on 2889 degrees of free
dom
## Multiple R-squared:  0.8637, Adjusted R-squared:  0.86
18
## F-statistic: 457.8 on 40 and 2889 DF,  p-value: < 2.2e
-16
```

8.4 6.4 Transformed Model Coefficient Table

```
transformed_coefficient_table <- broom::tidy(
  transformed_model,
  conf.int = TRUE,
  conf.level = 0.95
)

transformed_coefficient_table <- transformed_coefficient_
table[, c(
  "term",
  "estimate",
  "std.error",
  "statistic",
  "p.value",
  "conf.low",
  "conf.high"
)]

colnames(transformed_coefficient_table) <- c(
  "Variable",
  "Estimate",
  "Standard Error",
  "t Statistic",
  "p-value",
  "95% CI Lower",
  "95% CI Upper"
)

kable(
  transformed_coefficient_table,
```

```
  digits = 4,
  caption = "Coefficient Estimates for the Box-Cox Transf
ormed Model"
)
```

Coefficient Estimates for the Box-Cox Transformed Model

Variable	Estimate	Standard Error	t Statistic	p-value	95% CI Lower	95% CI Upper
(Intercept)	-31.0532	8.5501	-3.6319	0.0003	-47.8181	-14.2882
living_area	0.0049	0.0002	30.8271	0.0000	0.0046	0.0052
basement_area	0.0021	0.0002	11.8670	0.0000	0.0017	0.0024
overall_qualityPoor	3.8377	1.7748	2.1623	0.0307	0.3576	7.3177
overall_qualityFair	8.8908	1.6265	5.4663	0.0000	5.7016	12.0799
overall_qualityBelow_Average	11.4622	1.5681	7.3094	0.0000	8.3874	14.5369
overall_qualityAverage	13.6029	1.5606	8.7166	0.0000	10.5429	16.6628
overall_qualityAbove_Average	14.9645	1.5643	9.5663	0.0000	11.8973	18.0317
overall_qualityGood	16.4420	1.5726	10.4550	0.0000	13.3584	19.5256
overall_qualityVery_Good	18.8435	1.5895	11.8548	0.0000	15.7268	21.9602
overall_qualityExcellent	23.1071	1.6256	14.2147	0.0000	19.9197	26.2945
overall_qualityVery_Excellent	23.3101	1.7114	13.6203	0.0000	19.9544	26.6659
year_built	0.0410	0.0043	9.5303	0.0000	0.0326	0.0495
neighborhoodCollege_Creek	0.6965	0.2940	2.3689	0.0179	0.1200	1.2730
neighborhoodOld_Town	-1.5245	0.2977	-5.1212	0.0000	-2.1082	-0.9408
neighborhoodEdwards	-1.2796	0.2737	-4.6754	0.0000	-1.8163	-0.7430
neighborhoodSomerset	1.0559	0.3476	3.0375	0.0024	0.3743	1.7375
neighborhoodNorthridge_Heights	2.7815	0.3860	7.2059	0.0000	2.0246	3.5384
neighborhoodGilbert	0.2219	0.3381	0.6564	0.5116	-0.4410	0.8848
neighborhoodSawyer	-0.0494	0.2942	-0.1681	0.8666	-0.6262	0.5274
neighborhoodNorthwest_Ames	0.2183	0.3252	0.6711	0.5022	-0.4195	0.8560
neighborhoodSawyer_West	-0.2275	0.3454	-0.6589	0.5100	-0.9047	0.4496
neighborhoodMitchell	0.1702	0.3380	0.5035	0.6147	-0.4926	0.8330
neighborhoodBrookside	-0.2512	0.3536	-0.7105	0.4775	-0.9445	0.4421
neighborhoodCrawford	3.5943	0.3546	10.1353	0.0000	2.8989	4.2896
neighborhoodIowa_DOT_and_Rail_Road	-2.4378	0.3850	-6.3312	0.0000	-3.1928	-1.6828
neighborhoodTimberland	1.9831	0.4316	4.5947	0.0000	1.1368	2.8293
neighborhoodNorthridge	3.0425	0.4669	6.5166	0.0000	2.1271	3.9580
neighborhoodStone_Brook	3.1615	0.5250	6.0214	0.0000	2.1320	4.1910
neighborhoodSouth_and_West_of_Iowa_State_University	-0.9011	0.4924	-1.8299	0.0674	-1.8666	0.0644
neighborhoodClear_Creek	3.1197	0.4953	6.2992	0.0000	2.1486	4.0908
neighborhoodMeadow_Village	-3.4899	0.5590	-6.2431	0.0000	-4.5860	-2.3938
neighborhoodBriardale	-4.5144	0.5972	-7.5592	0.0000	-5.6853	-3.3434
neighborhoodBloomington_Heights	-0.4981	0.6432	-0.7744	0.4388	-1.7593	0.7631
neighborhoodVeenker	2.4052	0.6660	3.6114	0.0003	1.0993	3.7111
neighborhoodNorthpark_Villa	-2.2948	0.6717	-3.4163	0.0006	-3.6119	-0.9777
neighborhoodBlueste	-2.5451	1.0028	-2.5380	0.0112	-4.5114	-0.5789
neighborhoodGreens	0.2457	1.1347	0.2165	0.8286	-1.9793	2.4707
neighborhoodGreen_Hills	10.9210	2.2037	4.9557	0.0000	6.6000	15.2421
neighborhoodLandmark	-3.0616	3.1022	-0.9869	0.3238	-9.1443	3.0211
garage_capacity	1.2777	0.1065	12.0009	0.0000	1.0690	1.4865

8.5 6.5 Transformed Model Fit

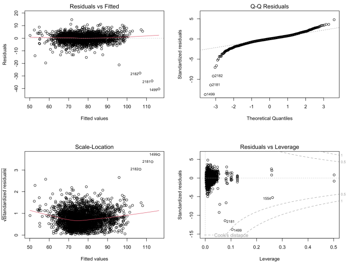
```r
transformed_model_summary <- summary(transformed_model)

n_transformed <- nobs(transformed_model)
k_transformed <- length(coef(transformed_model))

transformed_aic <- AIC(transformed_model)

transformed_aicc <- transformed_aic +
  (2 * k_transformed * (k_transformed + 1)) /
  (n_transformed - k_transformed - 1)

transformed_fit_table <- data.frame(
  Measure = c(
    "R-squared",
    "Adjusted R-squared",
    "Residual Standard Error",
    "AIC",
    "AICc",
    "BIC"
  ),
  Value = c(
    transformed_model_summary$r.squared,
    transformed_model_summary$adj.r.squared,
    transformed_model_summary$sigma,
    transformed_aic,
    transformed_aicc,
    BIC(transformed_model)
  )
)

kable(
  transformed_fit_table,
  digits = 4,
  caption = "Model Fit Measures for the Box-Cox Transformed Model"
)
```

Model Fit Measures for the Box-Cox Transformed Model

Measure	Value
R-squared	0.8637
Adjusted R-squared	0.8618
Residual Standard Error	3.0926
AIC	14973.6246
AICc	14974.8171
BIC	15224.9004

8.6 6.6 Standard Diagnostic Plots After Transformation

```r
par(mfrow = c(2, 2))

plot(transformed_model)
```

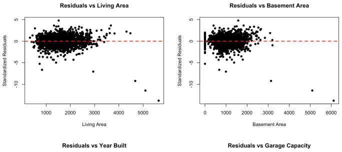


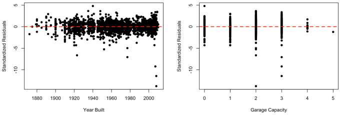
```r
par(mfrow = c(1, 1))
```

8.7 6.7 Residual Plots for Numerical

Predictors After Transformation

```r
transformed_standardized_residuals <- rstandard(transformed_model)
transformed_fitted_values <- fitted(transformed_model)

par(mfrow = c(2, 2))

plot(
  housing_clean$living_area,
  transformed_standardized_residuals,
  main = "Residuals vs Living Area",
  xlab = "Living Area",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(
  h = 0,
  col = "red",
  lty = 2,
  lwd = 2
)

plot(
  housing_clean$basement_area,
  transformed_standardized_residuals,
  main = "Residuals vs Basement Area",
  xlab = "Basement Area",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(
  h = 0,
  col = "red",
  lty = 2,
  lwd = 2
)

plot(
  housing_clean$year_built,
  transformed_standardized_residuals,
  main = "Residuals vs Year Built",
  xlab = "Year Built",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(
  h = 0,
  col = "red",
  lty = 2,
  lwd = 2
)

plot(
  housing_clean$garage_capacity,
  transformed_standardized_residuals,
  main = "Residuals vs Garage Capacity",
  xlab = "Garage Capacity",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(
  h = 0,
  col = "red",
  lty = 2,
  lwd = 2
)
```

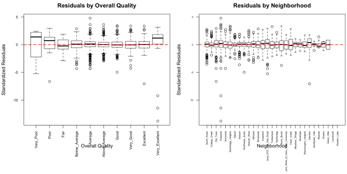




```
par(mfrow = c(1, 1))
```

8.8 6.8 Residual Plots for Categorical Predictors After Transformation

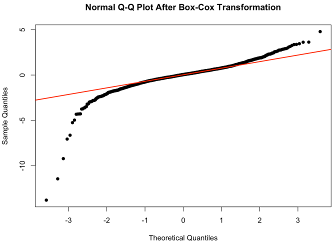
```
par(
  mfrow = c(1, 2),
  mar = c(8, 4, 3, 1)
)

boxplot(
  transformed_standardized_residuals ~
    housing_clean$overall_quality,
  main = "Residuals by Overall Quality",
  xlab = "Overall Quality",
  ylab = "Standardized Residuals",
  col = "white",
  las = 2,
  cex.axis = 0.75
)

abline(
  h = 0,
  col = "red",
  lty = 2,
  lwd = 2
)

boxplot(
  transformed_standardized_residuals ~
    housing_clean$neighborhood,
  main = "Residuals by Neighborhood",
  xlab = "Neighborhood",
  ylab = "Standardized Residuals",
  col = "white",
  las = 2,
  cex.axis = 0.5
)

abline(
  h = 0,
  col = "red",
  lty = 2,
  lwd = 2
)
```



```
par(mfrow = c(1, 1))
```

8.9 6.9 Normal Q-Q Plot After Transformation

```
qqnorm(
  transformed_standardized_residuals,
  main = "Normal Q-Q Plot After Box-Cox Transformation",
  pch = 16,
  col = "black"
)

qqline(
  transformed_standardized_residuals,
  col = "red",
  lwd = 2
)
```



9 7.1 Multicollinearity Assessment

```
vif_values <- car::vif(transformed_model)

vif_table <- data.frame(
  Variable = rownames(vif_values),
  GVIF = vif_values[, "GVIF"],
  Degrees_of_Freedom = vif_values[, "Df"],
  Adjusted_GVIF = vif_values[, "GVIF^(1/(2*Df))"],
  row.names = NULL
)

knitr::kable(
  vif_table,
  digits = 3,
  col.names = c(
    "Variable",
    "GVIF",
    "Degrees of Freedom",
    "Adjusted GVIF"
  ),
  caption = "Generalized Variance Inflation Factors for t
he Box-Cox Transformed Model"
)
```

Generalized Variance Inflation Factors for the Box-Cox Transformed Model

Variable	GVIF	Degrees of Freedom	Adjusted GVIF
living_area	1.986	1	1.409
basement_area	1.833	1	1.354
overall_quality	6.749	9	1.112
year_built	5.190	1	2.278
neighborhood	18.244	27	1.055
garage_capacity	2.011	1	1.418

9.1 7.2 Testing for a Quadratic Effect

```
# Fit a model that adds a quadratic term for Year Built
quadratic_model <- lm(
  sale_price_bc ~
    living_area +
    basement_area +
    overall_quality +
    year_built +
    I(year_built^2) +
    neighborhood +
    garage_capacity,
  data = housing_clean
)

# Display the model summary
summary(quadratic_model)
```

```
##
## Call:
## lm(formula = sale_price_bc ~ living_area + basement_ar
ea + overall_quality +
##     year_built + I(year_built^2) + neighborhood + gara
ge_capacity,
##     data = housing_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -40.321  -1.386   0.153   1.611  14.719
##
## Coefficients:
##
Estimate Std. Error
## (Intercept)                                         -
1.112e+02  3.459e+02
## living_area
4.914e-03  1.597e-04
## basement_area
2.083e-03  1.755e-04
## overall_qualityPoor
3.839e+00  1.775e+00
## overall_qualityFair
8.899e+00  1.627e+00
## overall_qualityBelow_Average
1.147e+01  1.569e+00
## overall_qualityAverage
1.361e+01  1.561e+00
## overall_qualityAbove_Average
1.498e+01  1.565e+00
## overall_qualityGood
1.646e+01  1.575e+00
## overall_qualityVery_Good
1.887e+01  1.593e+00
## overall_qualityExcellent
2.313e+01  1.630e+00
## overall_qualityVery_Excellent
2.334e+01  1.716e+00
## year_built
1.230e-01  3.540e-01
## I(year_built^2)                                     -
2.099e-05  9.060e-05
```

```
## neighborhoodCollege_Creek
7.284e-01  3.248e-01
## neighborhoodOld_Town                                -
1.498e+00  3.188e-01
## neighborhoodEdwards                                 -
1.268e+00  2.787e-01
## neighborhoodSomerset
1.095e+00  3.861e-01
## neighborhoodNorthridge_Heights
2.817e+00  4.160e-01
## neighborhoodGilbert
2.557e-01  3.682e-01
## neighborhoodSawyer                                  -
4.456e-02  2.950e-01
## neighborhoodNorthwest_Ames
2.197e-01  3.254e-01
## neighborhoodSawyer_West                             -
2.018e-01  3.628e-01
## neighborhoodMitchell
1.867e-01  3.455e-01
## neighborhoodBrookside                               -
2.412e-01  3.563e-01
## neighborhoodCrawford
3.601e+00  3.559e-01
## neighborhoodIowa_DOT_and_Rail_Road                  -
2.420e+00  3.931e-01
## neighborhoodTimberland
2.010e+00  4.467e-01
## neighborhoodNorthridge
3.058e+00  4.718e-01
## neighborhoodStone_Brook
3.186e+00  5.353e-01
## neighborhoodSouth_and_West_of_Iowa_State_University -
8.880e-01  4.957e-01
## neighborhoodClear_Creek
3.122e+00  4.955e-01
## neighborhoodMeadow_Village                          -
3.482e+00  5.601e-01
## neighborhoodBriardale                               -
4.510e+00  5.976e-01
## neighborhoodBloomington_Heights                     -
4.578e-01  6.664e-01
## neighborhoodVeenker
2.408e+00  6.662e-01
## neighborhoodNorthpark_Villa                         -
2.291e+00  6.720e-01
## neighborhoodBlueste                                 -
2.536e+00  1.004e+00
## neighborhoodGreens
2.416e-01  1.135e+00
## neighborhoodGreen_Hills
1.094e+01  2.206e+00
## neighborhoodLandmark                                -
3.034e+00  3.105e+00
## garage_capacity
1.279e+00  1.066e-01
##                                                     t
value Pr(>|t|)
## (Intercept)                                          -
0.321 0.747954
## living_area                                          3
0.768  < 2e-16 ***
## basement_area                                        1
1.867  < 2e-16 ***
## overall_qualityPoor
2.163 0.030630 *
## overall_qualityFair
5.469 4.91e-08 ***
## overall_qualityBelow_Average
7.311 3.40e-13 ***
## overall_qualityAverage
8.718  < 2e-16 ***
## overall_qualityAbove_Average
9.567  < 2e-16 ***
## overall_qualityGood                                  1
0.450  < 2e-16 ***
## overall_qualityVery_Good                             1
1.843  < 2e-16 ***
## overall_qualityExcellent                             1
4.194  < 2e-16 ***
## overall_qualityVery_Excellent                        1
3.600  < 2e-16 ***
## year_built
0.348 0.728221
## I(year_built^2)                                      -
0.232 0.816812
## neighborhoodCollege_Creek
2.243 0.024969 *
## neighborhoodOld_Town                                 -
4.699 2.73e-06 ***
## neighborhoodEdwards                                  -
4.549 5.62e-06 ***
## neighborhoodSomerset
2.835 0.004608 **
## neighborhoodNorthridge_Heights
6.772 1.53e-11 ***
## neighborhoodGilbert
0.694 0.487477
## neighborhoodSawyer                                   -
0.151 0.879918
## neighborhoodNorthwest_Ames
0.675 0.499565
## neighborhoodSawyer_West                              -
0.556 0.578044
## neighborhoodMitchell
0.540 0.588951
## neighborhoodBrookside                                -
0.677 0.498521
## neighborhoodCrawford                                 1
0.119  < 2e-16 ***
## neighborhoodIowa_DOT_and_Rail_Road                   -
6.156 8.50e-10 ***
## neighborhoodTimberland
4.499 7.11e-06 ***
## neighborhoodNorthridge
6.482 1.06e-10 ***
## neighborhoodStone_Brook
5.951 2.99e-09 ***
## neighborhoodSouth_and_West_of_Iowa_State_University  -
1.791 0.073322 .
## neighborhoodClear_Creek
6.302 3.39e-10 ***
## neighborhoodMeadow_Village                           -
6.217 5.79e-10 ***
## neighborhoodBriardale                                -
7.547 5.96e-14 ***
## neighborhoodBloomington_Heights                      -
0.687 0.492201
## neighborhoodVeenker
3.614 0.000306 ***
## neighborhoodNorthpark_Villa                          -
3.410 0.000660 ***
## neighborhoodBlueste                                  -
2.526 0.011579 *
## neighborhoodGreens
0.213 0.831430
## neighborhoodGreen_Hills
4.960 7.45e-07 ***
## neighborhoodLandmark                                 -
0.977 0.328555
## garage_capacity                                      1
1.999  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1
##
## Residual standard error: 3.093 on 2888 degrees of free
dom
## Multiple R-squared:  0.8637, Adjusted R-squared:  0.86
18
## F-statistic: 446.4 on 41 and 2888 DF,  p-value: < 2.2e
-16
```

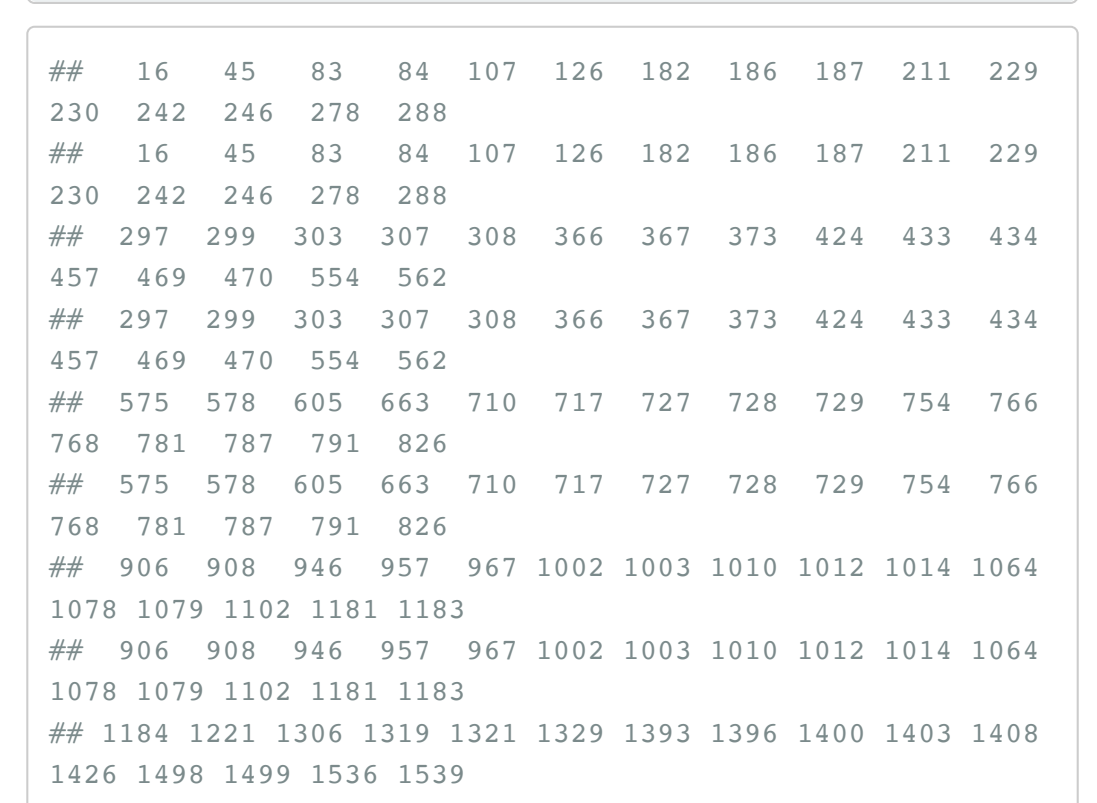
```r
# Compare the transformed model with the quadratic model
quadratic_test <- anova(
  transformed_model,
  quadratic_model
)

knitr::kable(
  quadratic_test,
  digits = 4,
  caption = "Partial F-test Comparing the Transformed Mod
el and the Quadratic Model"
)
```

Partial F-test Comparing the Transformed Model and the Quadratic Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
2889	27630.18	NA	NA	NA	NA
2888	27629.66	1	0.5135	0.0537	0.8168

9.2 7.3 Influence Diagnostics

```r
# Calculate Cook's distance for the transformed model
cook_values <- cooks.distance(transformed_model)

# Define the reference threshold
cook_threshold <- 4 / nrow(housing_clean)

# Plot Cook's distance
plot(
  cook_values,
  type = "h",
  main = "Cook's Distance for the Box-Cox Transformed Mod
el",
  xlab = "Observation",
  ylab = "Cook's Distance"
)

abline(
  h = cook_threshold,
  col = "red",
  lty = 2,
  lwd = 2
)
```



```r
# Identify observations above the Cook's distance thresho
ld
influential_cases <- which(
  cook_values > cook_threshold
)

# Display the number and indices of influential observati
ons
length(influential_cases)
```

```
## [1] 162
```

```r
influential_cases
```

```
##   16   45   83   84  107  126  182  186  187  211  229
230  242  246  278  288
##   16   45   83   84  107  126  182  186  187  211  229
230  242  246  278  288
##  297  299  303  307  308  366  367  373  424  433  434
457  469  470  554  562
##  297  299  303  307  308  366  367  373  424  433  434
457  469  470  554  562
##  575  578  605  663  710  717  727  728  729  754  766
768  781  787  791  826
##  575  578  605  663  710  717  727  728  729  754  766
768  781  787  791  826
##  906  908  946  957  967 1002 1003 1010 1012 1014 1064
1078 1079 1102 1181 1183
##  906  908  946  957  967 1002 1003 1010 1012 1014 1064
1078 1079 1102 1181 1183
## 1184 1221 1306 1319 1321 1329 1393 1396 1400 1403 1408
1426 1498 1499 1536 1539
## 1184 1221 1306 1319 1321 1329 1393 1396 1400 1403 1408
1426 1498 1499 1536 1539
## 1554 1556 1560 1569 1570 1580 1632 1638 1641 1642 1643
1694 1700 1702 1761 1768
## 1554 1556 1560 1569 1570 1580 1632 1638 1641 1642 1643
1694 1700 1702 1761 1768
## 1781 1783 1829 1857 1861 1902 1943 1946 1983 1994 2036
2044 2072 2075 2181 2182
## 1781 1783 1829 1857 1861 1902 1943 1946 1983 1994 2036
2044 2072 2075 2181 2182
## 2201 2210 2211 2228 2230 2240 2242 2246 2257 2299 2323
2327 2331 2333 2335 2342
## 2201 2210 2211 2228 2230 2240 2242 2246 2257 2299 2323
2327 2331 2333 2335 2342
## 2383 2421 2423 2430 2446 2451 2457 2522 2523 2596 2603
2656 2659 2667 2704 2731
## 2383 2421 2423 2430 2446 2451 2457 2522 2523 2596 2603
2656 2659 2667 2704 2731
## 2735 2736 2737 2738 2767 2830 2833 2844 2846 2858 2881
2884 2891 2892 2893 2894
## 2735 2736 2737 2738 2767 2830 2833 2844 2846 2858 2881
2884 2891 2892 2893 2894
## 2904 2908
## 2904 2908
```

```r
influence_summary <- data.frame(
  Measure = c(
    "Total observations",
    "Cook's distance threshold",
    "Observations above threshold",
    "Maximum Cook's distance"
  ),
  Value = c(
    nrow(housing_clean),
    cook_threshold,
    length(influential_cases),
    max(cook_values)
  )
)

knitr::kable(
  influence_summary,
  digits = 6,
  caption = "Summary of Cook's Distance Results"
)
```

Summary of Cook's Distance Results

Measure	Value
Total observations	2.930e+03
Cook's distance threshold	1.365e-03
Observations above threshold	1.620e+02
Maximum Cook's distance	NaN

```r
# Remove observations above the Cook's distance threshold
housing_final <- housing_clean[
  -influential_cases,
]

# Refit the model using the remaining observations
final_model <- lm(
  sale_price_bc ~
    living_area +
    basement_area +
    overall_quality +
    year_built +
    neighborhood +
    garage_capacity,
  data = housing_final
)

summary(final_model)
```

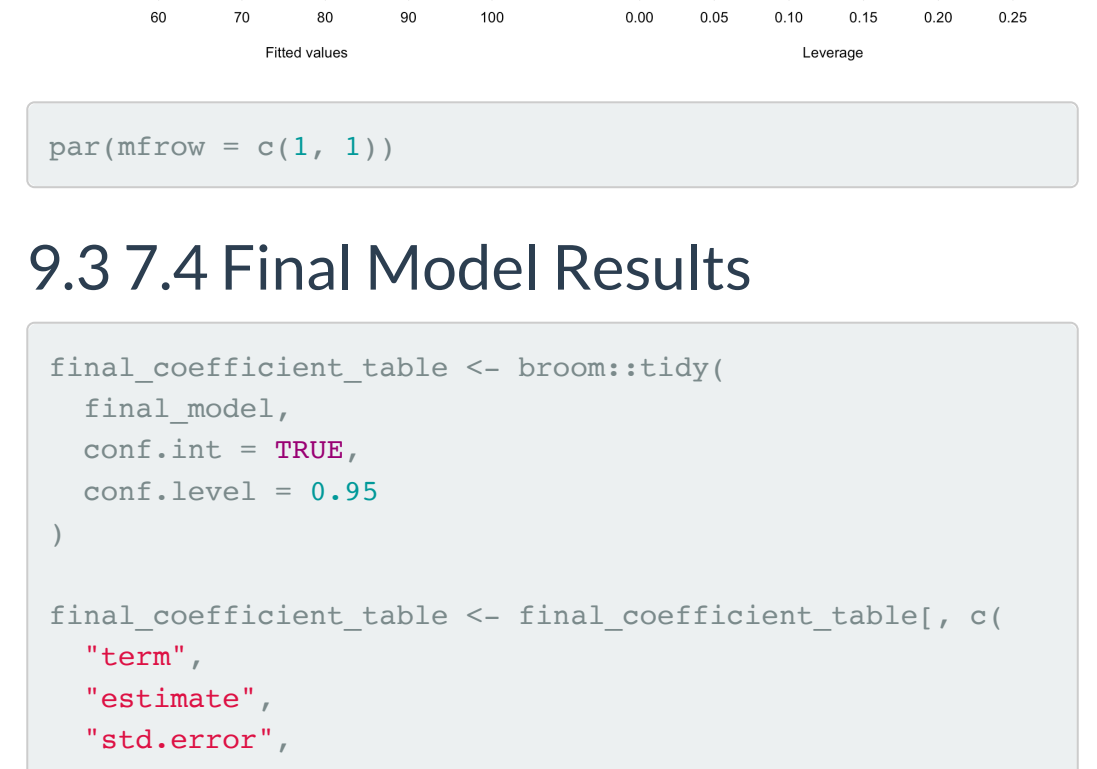
```
##
## Call:
## lm(formula = sale_price_bc ~ living_area + basement_ar
ea + overall_quality +
##     year_built + neighborhood + garage_capacity, data
= housing_final)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.4981 -1.2645  0.0947  1.5017  8.1486
##
## Coefficients:
##
Estimate Std. Error
## (Intercept)                                         -
3.709e+01  6.853e+00
## living_area
5.410e-03  1.291e-04
## basement_area
2.760e-03  1.432e-04
## overall_qualityFair
5.207e+00  1.231e+00
## overall_qualityBelow_Average
7.780e+00  1.172e+00
## overall_qualityAverage
9.711e+00  1.166e+00
## overall_qualityAbove_Average
1.094e+01  1.168e+00
## overall_qualityGood
1.234e+01  1.174e+00
## overall_qualityVery_Good
1.466e+01  1.188e+00
## overall_qualityExcellent
1.875e+01  1.217e+00
## overall_qualityVery_Excellent
1.984e+01  1.371e+00
## year_built
4.556e-02  3.433e-03
## neighborhoodCollege_Creek
5.375e-01  2.241e-01
## neighborhoodOld_Town                                -
1.180e+00  2.288e-01
## neighborhoodEdwards                                 -
8.697e-01  2.128e-01
## neighborhoodSomerset
8.948e-01  2.659e-01
## neighborhoodNorthridge_Heights
1.939e+00  3.023e-01
## neighborhoodGilbert
2.323e-01  2.578e-01
## neighborhoodSawyer
6.933e-02  2.212e-01
## neighborhoodNorthwest_Ames
1.361e-01  2.447e-01
## neighborhoodSawyer_West                             -
2.737e-01  2.640e-01
## neighborhoodMitchell
2.844e-01  2.584e-01
## neighborhoodBrookside                               -
9.478e-03  2.697e-01
## neighborhoodCrawford
3.547e+00  2.766e-01
## neighborhoodIowa_DOT_and_Rail_Road                  -
1.514e+00  3.039e-01
## neighborhoodTimberland
1.553e+00  3.358e-01
## neighborhoodNorthridge
2.212e+00  3.617e-01
## neighborhoodStone_Brook
2.256e+00  4.621e-01
## neighborhoodSouth_and_West_of_Iowa_State_University -
1.284e+00  3.865e-01
## neighborhoodClear_Creek
2.404e+00  4.153e-01
## neighborhoodMeadow_Village                          -
3.469e+00  4.182e-01
## neighborhoodBriardale                               -
4.195e+00  4.464e-01
## neighborhoodBloomington_Heights                     -
8.841e-01  5.234e-01
## neighborhoodVeenker
2.091e+00  6.150e-01
## neighborhoodNorthpark_Villa                         -
2.137e+00  5.013e-01
## neighborhoodBlueste                                 -
2.988e+00  8.319e-01
## neighborhoodGreens
1.228e+00  9.690e-01
## neighborhoodLandmark                                -
2.811e+00  2.313e+00
## garage_capacity
1.117e+00  8.291e-02
##                                                     t
value Pr(>|t|)
## (Intercept)                                          -
5.413 6.73e-08 ***
## living_area                                          4
1.916  < 2e-16 ***
## basement_area                                        1
9.280  < 2e-16 ***
## overall_qualityFair
4.228 2.43e-05 ***
## overall_qualityBelow_Average
6.636 3.87e-11 ***
## overall_qualityAverage
8.330  < 2e-16 ***
## overall_qualityAbove_Average
9.362  < 2e-16 ***
## overall_qualityGood                                  1
0.504  < 2e-16 ***
## overall_qualityVery_Good                             1
2.343  < 2e-16 ***
## overall_qualityExcellent                             1
5.403  < 2e-16 ***
## overall_qualityVery_Excellent                        1
4.468  < 2e-16 ***
## year_built                                           1
3.273  < 2e-16 ***
## neighborhoodCollege_Creek
2.399 0.016528 *
## neighborhoodOld_Town                                 -
5.155 2.72e-07 ***
## neighborhoodEdwards                                  -
4.087 4.49e-05 ***
## neighborhoodSomerset
3.365 0.000777 ***
## neighborhoodNorthridge_Heights
6.414 1.66e-10 ***
## neighborhoodGilbert
0.901 0.367495
## neighborhoodSawyer
0.313 0.754014
## neighborhoodNorthwest_Ames
0.556 0.578094
## neighborhoodSawyer_West                              -
1.037 0.299947
## neighborhoodMitchell
1.101 0.271152
## neighborhoodBrookside                                -
0.035 0.971965
## neighborhoodCrawford                                 1
2.822  < 2e-16 ***
## neighborhoodIowa_DOT_and_Rail_Road                   -
4.981 6.71e-07 ***
## neighborhoodTimberland
4.623 3.96e-06 ***
## neighborhoodNorthridge
6.117 1.09e-09 ***
## neighborhoodStone_Brook
4.883 1.11e-06 ***
## neighborhoodSouth_and_West_of_Iowa_State_University  -
3.323 0.000901 ***
## neighborhoodClear_Creek
5.788 7.91e-09 ***
## neighborhoodMeadow_Village                           -
8.295  < 2e-16 ***
## neighborhoodBriardale                                -
9.398  < 2e-16 ***
## neighborhoodBloomington_Heights                      -
1.689 0.091318 .
## neighborhoodVeenker
3.401 0.000682 ***
## neighborhoodNorthpark_Villa                          -
4.263 2.09e-05 ***
## neighborhoodBlueste                                  -
3.592 0.000334 ***
## neighborhoodGreens
1.268 0.205053
## neighborhoodLandmark                                 -
1.215 0.224283
## garage_capacity                                      1
3.477  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1
##
## Residual standard error: 2.305 on 2729 degrees of free
dom
## Multiple R-squared:  0.9085, Adjusted R-squared:  0.90
72
## F-statistic: 712.7 on 38 and 2729 DF,  p-value: < 2.2e
-16
```

```r
comparison_table <- data.frame(
  Model = c(
    "Transformed Model",
    "Model After Removing Influential Observations"
  ),
  Observations = c(
    nobs(transformed_model),
    nobs(final_model)
  ),
  R_squared = c(
    summary(transformed_model)$r.squared,
    summary(final_model)$r.squared
  ),
  Adjusted_R_squared = c(
    summary(transformed_model)$adj.r.squared,
    summary(final_model)$adj.r.squared
  ),
  Residual_Standard_Error = c(
    summary(transformed_model)$sigma,
    summary(final_model)$sigma
  ),
  AIC = c(
    AIC(transformed_model),
    AIC(final_model)
  ),
  BIC = c(
    BIC(transformed_model),
    BIC(final_model)
  )
)

knitr::kable(
  comparison_table,
  digits = 4,
  caption = "Comparison Before and After Removing Influen
tial Observations"
)
```

Comparison Before and After Removing Influential Observations

Model	Observations	R_squared	Adjusted_R_squared	Residual_Standard_Error	AIC	BIC
Transformed Model	2930	0.8637	0.8618	3.0926	14973.62	15224.90
Model After Removing Influential Observations	2768	0.9085	0.9072	2.3051	12519.31	12756.34

```r
par(mfrow = c(2, 2))

plot(final_model)
```



```r
par(mfrow = c(1, 1))
```

9.3 7.4 Final Model Results

```r
final_coefficient_table <- broom::tidy(
  final_model,
  conf.int = TRUE,
  conf.level = 0.95
)

final_coefficient_table <- final_coefficient_table[, c(
  "term",
  "estimate",
  "std.error",
  "statistic",
  "p.value",
  "conf.low",
  "conf.high"
)]

colnames(final_coefficient_table) <- c(
  "Variable",
  "Estimate",
  "Standard Error",
  "t Statistic",
  "p-value",
  "95% CI Lower",
  "95% CI Upper"
)

knitr::kable(
  final_coefficient_table,
  digits = 4,
  caption = "Coefficient Estimates for the Final Regressi
on Model"
)
```

Coefficient Estimates for the Final Regression Model

Variable	Estimate	Standard Error	t Statistic	p-value	95% CI Lower	95% CI Upper
(Intercept)	-37.0949	6.8528	-5.4131	0.0000	-50.5321	-23.6576
living_area	0.0054	0.0001	41.9165	0.0000	0.0052	0.0057
basement_area	0.0028	0.0001	19.2802	0.0000	0.0025	0.0030
overall_qualityFair	5.2070	1.2315	4.2283	0.0000	2.7923	7.6218
overall_qualityBelow_Average	7.7802	1.1724	6.6362	0.0000	5.4814	10.0791
overall_qualityAverage	9.7108	1.1658	8.3301	0.0000	7.4250	11.9967
overall_qualityAbove_Average	10.9388	1.1684	9.3620	0.0000	8.6477	13.2299
overall_qualityGood	12.3350	1.1743	10.5042	0.0000	10.0324	14.6376
overall_qualityVery_Good	14.6577	1.1875	12.3428	0.0000	12.3291	16.9863
overall_qualityExcellent	18.7491	1.2173	15.4027	0.0000	16.3623	21.1360
overall_qualityVery_Excellent	19.8398	1.3713	14.4682	0.0000	17.1510	22.5286
year_built	0.0456	0.0034	13.2731	0.0000	0.0388	0.0523
neighborhoodCollege_Creek	0.5375	0.2241	2.3985	0.0165	0.0981	0.9770
neighborhoodOld_Town	-1.1798	0.2288	-5.1552	0.0000	-1.6285	-0.7310
neighborhoodEdwards	-0.8697	0.2128	-4.0875	0.0000	-1.2869	-0.4525
neighborhoodSomerset	0.8948	0.2659	3.3648	0.0008	0.3734	1.4163
neighborhoodNorthridge_Heights	1.9392	0.3023	6.4144	0.0000	1.3464	2.5321
neighborhoodGilbert	0.2323	0.2578	0.9013	0.3675	-0.2731	0.7378
neighborhoodSawyer	0.0693	0.2212	0.3134	0.7540	-0.3645	0.5031
neighborhoodNorthwest_Ames	0.1361	0.2447	0.5562	0.5781	-0.3436	0.6158
neighborhoodSawyer_West	-0.2737	0.2640	-1.0367	0.2999	-0.7914	0.2440
neighborhoodMitchell	0.2844	0.2584	1.1006	0.2712	-0.2223	0.7910
neighborhoodBrookside	-0.0095	0.2697	-0.0351	0.9720	-0.5383	0.5193
neighborhoodCrawford	3.5470	0.2766	12.8224	0.0000	3.0046	4.0894
neighborhoodIowa_DOT_and_Rail_Road	-1.5139	0.3039	-4.9812	0.0000	-2.1099	-0.9180
neighborhoodTimberland	1.5526	0.3358	4.6230	0.0000	0.8941	2.2111
neighborhoodNorthridge	2.2122	0.3617	6.1168	0.0000	1.5030	2.9213
neighborhoodStone_Brook	2.2561	0.4621	4.8826	0.0000	1.3501	3.1621
neighborhoodSouth_and_West_of_Iowa_State_University	-1.2844	0.3865	-3.3234	0.0009	-2.0423	-0.5266
neighborhoodClear_Creek	2.4040	0.4153	5.7885	0.0000	1.5896	3.2183
neighborhoodMeadow_Village	-3.4694	0.4182	-8.2953	0.0000	-4.2895	-2.6493
neighborhoodBriardale	-4.1951	0.4464	-9.3976	0.0000	-5.0704	-3.3198
neighborhoodBloomington_Heights	-0.8841	0.5234	-1.6891	0.0913	-1.9104	0.1422
neighborhoodVeenker	2.0913	0.6150	3.4006	0.0007	0.8854	3.2973
neighborhoodNorthpark_Villa	-2.1368	0.5013	-4.2626	0.0000	-3.1197	-1.1538
neighborhoodBlueste	-2.9882	0.8319	-3.5922	0.0003	-4.6194	-1.3571
neighborhoodGreens	1.2284	0.9690	1.2676	0.2051	-0.6718	3.1285
neighborhoodLandmark	-2.8112	2.3128	-1.2155	0.2243	-7.3461	1.7238
garage_capacity	1.1173	0.0829	13.4772	0.0000	0.9548	1.2799

```r
final_glance <- broom::glance(final_model)

final_fit_table <- data.frame(
  Measure = c(
    "Number of Observations",
    "R-squared",
    "Adjusted R-squared",
    "Residual Standard Error",
    "F Statistic",
    "Model p-value",
    "AIC",
    "BIC"
  ),
  Value = c(
    nobs(final_model),
    final_glance$r.squared,
    final_glance$adj.r.squared,
    final_glance$sigma,
    final_glance$statistic,
    final_glance$p.value,
    AIC(final_model),
    BIC(final_model)
  )
)

knitr::kable(
  final_fit_table,
  digits = 4,
  caption = "Model Fit Measures for the Final Regression Model"
)
```

Model Fit Measures for the Final Regression Model

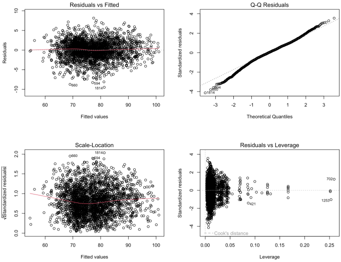
Measure	Value
Number of Observations	2768.0000
R-squared	0.9085
Adjusted R-squared	0.9072
Residual Standard Error	2.3051
F Statistic	712.6644
Model p-value	0.0000
AIC	12519.3073
BIC	12756.3425

```
formula(final_model)
```

```
## sale_price_bc ~ living_area + basement_area + overall_quality +
##     year_built + neighborhood + garage_capacity
```

¥

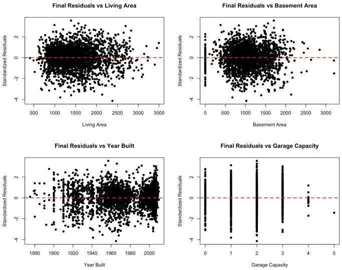
```
par(mfrow = c(2, 2))

plot(final_model)
```



```
par(mfrow = c(1, 1))
```

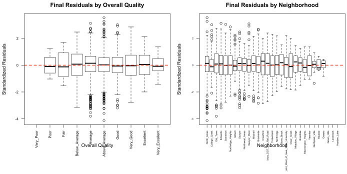
```
final_standardized_residuals <- rstandard(final_model)

par(mfrow = c(2, 2))

plot(
  housing_final$living_area,
  final_standardized_residuals,
  main = "Final Residuals vs Living Area",
  xlab = "Living Area",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)

plot(
  housing_final$basement_area,
  final_standardized_residuals,
  main = "Final Residuals vs Basement Area",
  xlab = "Basement Area",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)

plot(
```

```
  housing_final$year_built,
  final_standardized_residuals,
  main = "Final Residuals vs Year Built",
  xlab = "Year Built",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)

plot(
  housing_final$garage_capacity,
  final_standardized_residuals,
  main = "Final Residuals vs Garage Capacity",
  xlab = "Garage Capacity",
  ylab = "Standardized Residuals",
  pch = 16,
  col = "black"
)
abline(h = 0, col = "red", lty = 2, lwd = 2)
```



```
par(mfrow = c(1, 1))
```

```
par(mfrow = c(1, 2), mar = c(8, 4, 3, 1))

boxplot(
  final_standardized_residuals ~ housing_final$overall_qu
ality,
  main = "Final Residuals by Overall Quality",
  xlab = "Overall Quality",
  ylab = "Standardized Residuals",
  col = "white",
  las = 2,
  cex.axis = 0.75
)

abline(h = 0, col = "red", lty = 2, lwd = 2)

boxplot(
  final_standardized_residuals ~ housing_final$neighborho
od,
  main = "Final Residuals by Neighborhood",
  xlab = "Neighborhood",
  ylab = "Standardized Residuals",
  col = "white",
  las = 2,
  cex.axis = 0.5
)

abline(h = 0, col = "red", lty = 2, lwd = 2)
```



```
par(mfrow = c(1, 1))
```

10 8. Model Validation

10.1 8.1 Training and Test Set Split

```
set.seed(302)

train_index <- sample(
  seq_len(nrow(housing_final)),
  size = floor(0.8 * nrow(housing_final))
)

training_data <- housing_final[train_index, ]
test_data <- housing_final[-train_index, ]

split_table <- data.frame(
  Dataset = c("Training Set", "Test Set"),
  Observations = c(
    nrow(training_data),
    nrow(test_data)
  )
)

knitr::kable(
  split_table,
  caption = "Sample Sizes for the Training and Test Sets"
)
```

Sample Sizes for the Training and Test Sets

Dataset	Observations
Training Set	2214
Test Set	554

10.2 8.2 Fit the Validation Model

```
validation_model <- lm(
  sale_price_bc ~
    living_area +
    basement_area +
    overall_quality +
    year_built +
    neighborhood +
    garage_capacity,
  data = training_data
)

summary(validation_model)
```

```
##
## Call:
## lm(formula = sale_price_bc ~ living_area + basement_ar
ea + overall_quality +
##     year_built + neighborhood + garage_capacity, data
= training_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.4357 -1.3100  0.1344  1.4788  8.2120
##
## Coefficients:
##
Estimate Std. Error
## (Intercept)                                         -
3.579e+01  7.867e+00
## living_area
5.342e-03  1.446e-04
## basement_area
2.653e-03  1.591e-04
## overall_qualityFair
5.379e+00  1.710e+00
## overall_qualityBelow_Average
7.860e+00  1.663e+00
## overall_qualityAverage
9.886e+00  1.658e+00
## overall_qualityAbove_Average
1.107e+01  1.660e+00
## overall_qualityGood
1.245e+01  1.665e+00
```

```
## overall_qualityVery_Good
1.475e+01  1.678e+00
## overall_qualityExcellent
1.904e+01  1.706e+00
## overall_qualityVery_Excellent
1.980e+01  1.830e+00
## year_built
4.495e-02  3.892e-03
## neighborhoodCollege_Creek
5.218e-01  2.546e-01
## neighborhoodOld_Town                                -
1.418e+00  2.592e-01
## neighborhoodEdwards                                 -
9.034e-01  2.430e-01
## neighborhoodSomerset
9.974e-01  3.002e-01
## neighborhoodNorthridge_Heights
2.113e+00  3.401e-01
## neighborhoodGilbert
1.913e-01  2.898e-01
## neighborhoodSawyer
8.449e-03  2.470e-01
## neighborhoodNorthwest_Ames
7.074e-02  2.747e-01
## neighborhoodSawyer_West                             -
4.354e-01  2.953e-01
## neighborhoodMitchell
7.744e-02  2.856e-01
## neighborhoodBrookside                               -
1.885e-01  3.033e-01
## neighborhoodCrawford
3.591e+00  3.121e-01
## neighborhoodIowa_DOT_and_Rail_Road                  -
1.809e+00  3.343e-01
## neighborhoodTimberland
1.788e+00  3.766e-01
## neighborhoodNorthridge
2.451e+00  4.108e-01
## neighborhoodStone_Brook
2.295e+00  5.377e-01
## neighborhoodSouth_and_West_of_Iowa_State_University -
1.187e+00  4.217e-01
## neighborhoodClear_Creek
2.297e+00  4.694e-01
## neighborhoodMeadow_Village                          -
3.490e+00  4.687e-01
## neighborhoodBriardale                               -
4.042e+00  5.108e-01
## neighborhoodBloomington_Heights                     -
7.830e-01  6.083e-01
## neighborhoodVeenker
2.082e+00  6.648e-01
## neighborhoodNorthpark_Villa                         -
2.194e+00  5.551e-01
## neighborhoodBlueste                                 -
2.969e+00  8.954e-01
## neighborhoodGreens
1.091e+00  1.070e+00
## neighborhoodLandmark                                -
2.846e+00  2.327e+00
## garage_capacity
1.105e+00  9.374e-02
##                                                     t
value Pr(>|t|)
## (Intercept)                                          -
4.549 5.69e-06 ***
## living_area                                          3
6.947  < 2e-16 ***
## basement_area                                        1
6.677  < 2e-16 ***
## overall_qualityFair
3.145 0.001683 **
## overall_qualityBelow_Average
4.727 2.42e-06 ***
## overall_qualityAverage
5.965 2.86e-09 ***
## overall_qualityAbove_Average
6.669 3.26e-11 ***
## overall_qualityGood
7.476 1.10e-13 ***
## overall_qualityVery_Good
8.787  < 2e-16 ***
## overall_qualityExcellent                             1
1.160  < 2e-16 ***
## overall_qualityVery_Excellent                        1
0.818  < 2e-16 ***
```

```
## year_built                                           1
1.550  < 2e-16 ***
## neighborhoodCollege_Creek
2.049 0.040549 *
## neighborhoodOld_Town                                 -
5.471 4.98e-08 ***
## neighborhoodEdwards                                  -
3.717 0.000207 ***
## neighborhoodSomerset
3.323 0.000906 ***
## neighborhoodNorthridge_Heights
6.213 6.20e-10 ***
## neighborhoodGilbert
0.660 0.509163
## neighborhoodSawyer
0.034 0.972711
## neighborhoodNorthwest_Ames
0.258 0.796797
## neighborhoodSawyer_West                              -
1.475 0.140486
## neighborhoodMitchell
0.271 0.786283
## neighborhoodBrookside                                -
0.622 0.534268
## neighborhoodCrawford                                 1
1.507  < 2e-16 ***
## neighborhoodIowa_DOT_and_Rail_Road                   -
5.412 6.90e-08 ***
## neighborhoodTimberland
4.749 2.17e-06 ***
## neighborhoodNorthridge
5.967 2.82e-09 ***
## neighborhoodStone_Brook
4.267 2.07e-05 ***
## neighborhoodSouth_and_West_of_Iowa_State_University  -
2.815 0.004920 **
## neighborhoodClear_Creek
4.893 1.07e-06 ***
## neighborhoodMeadow_Village                           -
7.447 1.37e-13 ***
## neighborhoodBriardale                                -
7.915 3.91e-15 ***
## neighborhoodBloomington_Heights                      -
1.287 0.198195
## neighborhoodVeenker
3.132 0.001760 **
## neighborhoodNorthpark_Villa                          -
3.952 8.01e-05 ***
## neighborhoodBlueste                                  -
3.316 0.000928 ***
## neighborhoodGreens
1.020 0.307722
## neighborhoodLandmark                                 -
1.223 0.221429
## garage_capacity                                      1
1.791  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1
##
## Residual standard error: 2.317 on 2175 degrees of free
dom
## Multiple R-squared:  0.9089, Adjusted R-squared:  0.90
74
## F-statistic: 571.4 on 38 and 2175 DF,  p-value: < 2.2e
-16
```

10.3 8.3 Test-Set Performance

```r
# Predict the transformed response for the test set
test_predictions_bc <- predict(
  validation_model,
  newdata = test_data
)

# Actual transformed response
test_actual_bc <- test_data$sale_price_bc

# Calculate prediction accuracy
test_mse <- mean(
  (test_actual_bc - test_predictions_bc)^2
)

test_rmse <- sqrt(test_mse)

test_mae <- mean(
  abs(test_actual_bc - test_predictions_bc)
)

test_r_squared <- 1 -
  sum((test_actual_bc - test_predictions_bc)^2) /
  sum((test_actual_bc - mean(test_actual_bc))^2)

validation_table <- data.frame(
  Measure = c(
    "Test MSE",
    "Test RMSE",
    "Test MAE",
    "Test R-squared"
  ),
  Value = c(
    test_mse,
    test_rmse,
    test_mae,
    test_r_squared
  )
)

knitr::kable(
  validation_table,
  digits = 4,
  caption = "Test-Set Performance of the Final Regression Model"
)
```

Test-Set Performance of the Final Regression Model

Measure	Value
Test MSE	5.1710
Test RMSE	2.2740
Test MAE	1.7548
Test R-squared	0.9048

10.4 8.4 Performance on the Original Sale Price Scale

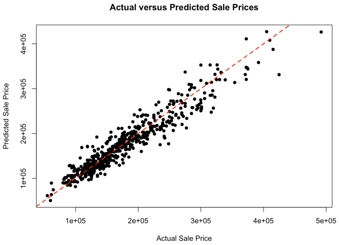
```r
lambda_selected <- 0.25

# Convert predictions back to the original sale-price scale
test_predictions_price <- (
  lambda_selected * test_predictions_bc + 1
)^(1 / lambda_selected)

test_actual_price <- test_data$sale_price

price_mse <- mean(
  (test_actual_price - test_predictions_price)^2
)

price_rmse <- sqrt(price_mse)

price_mae <- mean(
  abs(test_actual_price - test_predictions_price)
)

price_r_squared <- 1 -
  sum((test_actual_price - test_predictions_price)^2) /
  sum((test_actual_price - mean(test_actual_price))^2)
```

```
original_scale_table <- data.frame(
  Measure = c(
    "Test RMSE in Dollars",
    "Test MAE in Dollars",
    "Test R-squared"
  ),
  Value = c(
    price_rmse,
    price_mae,
    price_r_squared
  )
)

knitr::kable(
  original_scale_table,
  digits = 2,
  caption = "Test-Set Performance on the Original Sale Price Scale"
)
```

Test-Set Performance on the Original Sale Price Scale

Measure	Value
Test RMSE in Dollars	20701.55
Test MAE in Dollars	15348.43
Test R-squared	0.91

10.5 8.5 Actual and Predicted Sale Prices

```
plot(
  test_actual_price,
  test_predictions_price,
  main = "Actual versus Predicted Sale Prices",
  xlab = "Actual Sale Price",
  ylab = "Predicted Sale Price",
  pch = 16,
  col = "black"
)

abline(
  a = 0,
  b = 1,
  col = "red",
  lwd = 2,
  lty = 2
)
```